

João André Roque Costa

MSC GRADUATE, COMPUTER SCIENCE AND ENGINEERING

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Junior AI Developer with a Master's degree in Computer Science (AI and Data Science) from **Instituto Superior Técnico**. Experienced in developing **AI agent solutions**, process automation, and managing **Azure** resource groups. Skilled in Python, Azure Foundry, Azure Blob Storage, MongoDB, and data analysis techniques.

Professional Experience

Deloitte

Coimbra, Portugal

Tech Consultant

Apr. 2026 - Present

- Developed a multi-agent application for data migration and generation tasks (ClaudeSDK, Azure Foundry, Azure Blob Storage and MongoDB).
- Ported and updated an Azure Resource Group between two subscriptions (Azure Managed Instances, Data Factory, Databricks, ...).
- Automated an HR report generation process using Python and Playright.

Instituto Superior Técnico

Lisbon, Portugal

Invited Teaching Assistant

Oct. 2024 - Jul. 2025

- Lectured and evaluated practical classes on Databases and Data Science.
- Assisted students with hands-on projects in data treatment, visualization, and analysis techniques using Python (pandas/matplotlib).

Education

Instituto Superior Técnico

Lisbon, Portugal

M.Sc. Computer Science and Engineering

Sep 2023 - Nov 2025

- Specialized in **Data Science** with emphasis in **Artificial Intelligence**, maintaining a focus on scalable data architectures and distributed systems.
- Relevant Coursework: Information Visualization, Data and Information Systems Management, Data Analysis and Integration, Machine Learning, Deep Learning, Cloud Computing and Virtualization, Optimization Algorithms.
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM).
 - Developed a novel extension to the MVU algorithm that enables nonlinear dimensionality reduction on disconnected datasets.
 - Engineered a decomposition-based approach that allows for parallel processing, significantly improving scalability for large-scale datasets that were previously computationally infeasible.
 - Achieved equal-quality embeddings and preservation of local geometric structures compared to the standard method.

Instituto Superior Técnico

Lisbon, Portugal

B.Sc. Computer Science and Engineering

Sep 2020 - Jul 2023

- Developed a strong foundation in **Distributed Systems**, **Operating Systems**, and **Database Theory**.
- Collaborated on several full-stack and systems-level projects, emphasizing memory management in C/C++ and relational schema design in SQL.

Extracurricular

Personal Software Projects (Self-Hosted)

- Engineered an automated betting bot for parimutuel systems using a modified Kelly Criterion for risk management and optimal stake sizing.
- Developed a custom Discord-based ELO ranking and automatic matchmaking system.
- Architected a multi-platform arbitrage engine and inventory manager. With automated price monitoring and trade execution across multiple marketplaces.
- Built a full-stack data visualization dashboard to analyze player performance metrics, converting raw API data into actionable strategic insights.

Skills & Competencies

Programming	Python (Claude-Agent-SDK, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Playright), Java, C++, C#
Databases	SQL, NoSQL, Microsoft SQL Server (SSMS), PostgreSQL, MongoDB, MySQL, pgAdmin
Data Engineering	Pentaho PDI, SQL Profiler, Tuning Advisor (DTA), DataCleaner tool
Business Intelligence	Pentaho (Server, Schema Workbench, Report Designer), Saiku Analytics
Cloud & Architecture	AWS, Azure, ArchiMate, Camunda Modeler, Enterprise Architecture (Sparx)
Developer Tools	Git, Linux, Bash Scripting, RESTful APIs, LaTeX
Languages	Portuguese (Native), English (Fluent)